Auditing SHAP Explanations Against Latent Failure Mechanisms in Predictive Maintenance

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***Abstract*—Machine learning models predict equipment failures in industrial systems, but whether their explanations reflect the physical failure mechanisms is rarely tested statistically. We propose an audit that measures, for each known failure mechanism, how much SHAP attribution falls on the mechanism's trigger features, how specific that concentration is to the mechanism, and whether any other feature takes unusual attribution. Explanations are computed out of sample, and each statistic is compared with up to 1,000 models retrained on shuffled labels. On the AI4I 2020 dataset, a tuned XGBoost classifier (test F1 = 0.89) shows significant alignment for three of four physical mechanisms across five seeds, against a null re-tuned on every permutation, with a random forest, and with interventional TreeSHAP; a neural network explained by Integrated Gradients or KernelSHAP shows weaker alignment. The tool-wear verdict depends on the SHAP formulation, which we trace mainly to the reference distribution: against normal operation, or with conditional Shapley values of feature groups, tool wear is also aligned. A planted shortcut sensor shows that models can keep high accuracy while relying on a misleading feature, that weak shortcuts escape detection, and that at 300,000 observations only the shortcut statistic still detects a strong one. On measured bearing vibration and Tennessee Eastman process data, the audit confirms mechanism-consistent attribution for inner- and outer-race faults and a reactor-cooling fault, flags an injected acquisition-gain shortcut that leaves accuracy perfect, and flags a process fault whose effect spreads beyond its expected features. Predictive accuracy alone therefore cannot establish the physical relevance of explanations.**

***Index Terms*—Explainable AI, SHAP, predictive maintenance, feature attribution evaluation, physical failure mechanisms**

I. Introduction

Predictive maintenance uses machine-condition data to anticipate faults and reduce unplanned downtime [1]. Its models range from decision trees to boosted ensembles and neural networks [2], [3]. Knowing which sensor readings drive a prediction helps maintenance teams assess an alarm, and explanation methods such as LIME [2], [4] and SHAP [5], [6] are now common in the field, including studies of how preprocessing and model complexity affect explanation quality [7].

High predictive performance can coexist with reliance on shortcuts or correlations unrelated to the intended task [8], [9]. In predictive maintenance, this raises the question of whether explanations emphasize the expected features for each failure type. SHAP assigns feature contributions to individual predictions [10], but these contributions do not by themselves establish the physical relevance of those inputs [11]. Evaluating attributions needs a ground-truth reference [12]: here, which mechanism occurred and which features its rule uses, together with a statistical reference such as label permutation [13].

We use the AI4I 2020 dataset [14], whose recorded failure types and documented generation rules provide such a reference, as well as CWRU bearing data, whose fault locations have known defect frequencies, and Tennessee Eastman process data, whose faults act on known cooling loops. Prior AI4I work assessed explanations on selected examples [15], global SHAP rankings [16], or SHAP–LIME agreement [17]; none tests mechanism-level attribution on held-out data against a statistical reference, or whether such a test detects reliance on a fabricated feature.

Our audit trains a classifier on the overall failure label only and uses the recorded failure types to group its SHAP explanations by mechanism (Fig. 1). Three statistics measure the attribution share on a mechanism's expected trigger features, how much larger that share is than on other failures, and the largest excess share taken by any other feature. Explanations come from five-fold cross-fitting, so no observation is explained by a model trained on it, and each statistic is compared with pipelines retrained on shuffled labels, with Holm correction across mechanisms.

Our contributions are as follows:

- A mechanism-level SHAP audit with held-out explanations and a retrained shuffled-label null; we show that a random-subset baseline can approve a model trained on meaningless labels.
- A robustness study across seeds, features, labels, imbalance treatments, a re-tuned null, two tree ensembles, an MLP with four explainers, and dependence-aware SHAP, locating the tool-wear disagreement in the reference distribution.
- A planted-shortcut study showing that high accuracy can conceal shortcut reliance and that weak shortcuts escape detection.
- A scaling study up to 300,000 observations (runtime up to one million), showing that only the shortcut statistic catches a strong shortcut at the largest size.

- Studies on measured CWRU bearing vibration and Tennessee Eastman process data, whose expected features follow from bearing kinematics and process structure, including an acquisition shortcut.
- A benchmark of seven classifiers with confidence intervals, and released code, models, per-observation attributions, and reproduction scripts.

II. Related Work and Background

Surveys of explainable predictive maintenance cover approaches from inherently interpretable models to surrogate models, LIME, and SHAP [1], [18]. Our audit sits between local and global explanation, aggregating local attributions within each recorded failure mechanism.

Shapley values have supported breakdown diagnosis in petrochemical plants [5], and LIME, SHAP, and attribute-wise explanations have been compared on PHM08-CMAPSS [3]. Closest to our setting, Matzka [15] assessed explanation quality on AI4I 2020 against its known failure mechanisms using five examples from each of four physical failure types. Later work interpreted SHAP rankings on the same data [16], and Obeten and Jimoh measured SHAP–LIME agreement [17]. We extend this direction from selected examples and method agreement to a statistical test over all held-out observations.

SHAP assigns each feature an additive contribution that, together with a baseline value, reconstructs the explained output [10]. Covert et al. [19] show that how removal-based explanations represent unavailable features is part of the explanation's definition. Shapley variants condition on the remaining features, marginalize over a background, or substitute a single baseline [20], and these choices answer different questions about the model and the data [21], [22], particularly for correlated inputs [23], [24]. For tree ensembles, path-dependent TreeSHAP weights each path by its training cover, whereas interventional TreeSHAP averages over an explicit background sample [25]. Faster exact algorithms, such as Linear TreeSHAP [26] and the quadrature-based Quadrature-TreeSHAP [27], accelerate the path-dependent computation while retaining its formulation. The choice matters here because engineered features share information with the measurements they are built from, and Section V-D shows that it can decide a mechanism's verdict; agreement with trigger features must therefore be read in light of the explanation formulation, not as proof of physical causation.

Benchmarks with known ground truth, such as XAI-Bench [28], XAI-Units [29], BAM [30], and CLEVR-XAI [31], test whether attributions recover planted relevance; OpenXAI evaluates faithfulness, stability, and fairness [32], and soundness and completeness metrics weigh attribution values rather than ranks [33]; and ROAR [34] and its retraining-free successor ROAD [35] remove highly attributed inputs and measure the change in performance. Global sensitivity analysis based on the Hilbert–Schmidt independence criterion (HSIC) measures statistical dependence between inputs and outputs without a removal step [36]. These characterize the model or explainer; agreement with a failure mechanism additionally needs an external reference, such as documented failure rules.

Randomization provides the reference we use. Label-permutation tests assess classifier performance [37], and outcome permutations calibrate feature-importance measures [13]. Some visually convincing saliency methods are insensitive to randomized parameters or labels [38] or unreliable under simple input transformations [39], and post-hoc explanations such as LIME and SHAP can be deliberately manipulated [40], so explanations need explicit controls. Research on shortcut learning gives a further reason: models can exploit predictive associations that do not represent the intended mechanism [8], [9].

III. Methodology

A. Problem Definition

A model may predict a failure correctly without relying on the conditions that caused it. We therefore ask: *when a particular failure occurs, does the model attribute more importance to the features in that failure's physical rule?*

Let $\mathcal{D} = \{(x_i, y_i, z_i)\}_{i=1}^{N}$ denote a dataset where $x_i \in \mathbb{R}^d$ contains the input features, $y_i \in \{0, 1\}$ indicates overall equipment failure, and $z_i \in \{0, 1\}^M$ records M individual failure mechanisms. A classifier f is trained using only (x_i, y_i); the mechanism indicators are reserved for evaluating its explanations.

For each mechanism m, domain knowledge identifies a set E_m of *trigger features* involved in its physical rule. We define $S_m = \{i : z_{i,m} = 1\}$ as the observations where mechanism m occurs and $F = \{i : y_i = 1\}$ as all observations labeled as failures. Let $\phi_j(x_i)$ denote the SHAP attribution of feature j for observation i.

The audit tests whether attribution concentrates on the expected trigger features, whether that concentration is specific to the mechanism, and whether unrelated features receive unusual attribution (Fig. 1). It measures agreement with known failure rules, not physical causation.

B. Dataset and Failure Mechanisms

AI4I 2020 [15] is a synthetic industrial predictive maintenance dataset representing a milling process. It contains 10,000 observations, each recording product quality (`L`, `M`, or `H`), air and process temperatures, rotational speed, torque, tool wear, an overall failure label, and five failure-mechanism indicators. Only 339 observations (3.4%) are labeled as failures, making the prediction task highly imbalanced.

The dataset describes five mechanisms: tool wear failure (TWF), heat dissipation failure (HDF), power failure (PWF), overstrain failure (OSF), and random failure (RNF). The overall failure label is intended to follow the logical OR of these mechanisms:

$$y_i^{\mathrm{OR}} = \mathbb{I}\left(\sum_{m=1}^{M} z_{i,m} \geq 1\right), \tag{1}$$

where $\mathbb{I}(\cdot)$ is the indicator function. Table I lists the documented rules and the trigger features used in the audit.

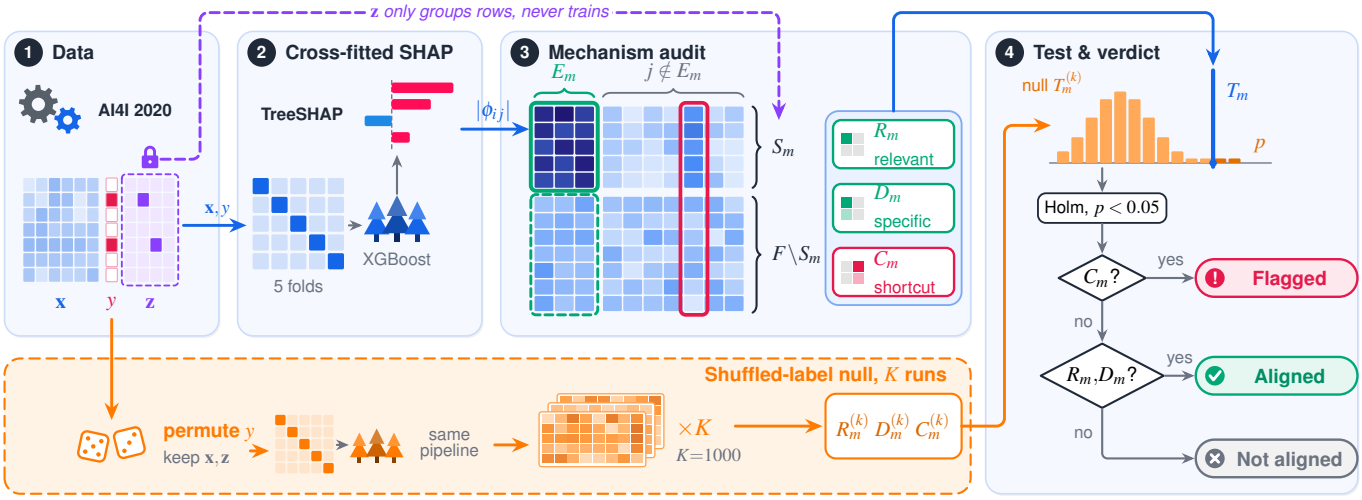


Fig. 1. Mechanism-level SHAP audit. (1) The classifier is trained on $\mathbf{x}$ and the failure label y; the mechanism flags $\mathbf{z}$ only group observations. (2) Five-fold cross-fitting gives every observation a TreeSHAP explanation from a model that did not train on it. (3) The absolute attributions $|\phi_{ij}|$ are split into mechanism observations S_m and other failures $F \setminus S_m$, and into trigger features E_m and all other features, giving the relevant share R_m, the specificity D_m, and the shortcut score C_m. Green marks the trigger features E_m (solid: S_m rows, giving R_m; solid versus dashed gives D_m); red marks the non-trigger feature with the largest excess, giving C_m. (4) Each observed statistic T_m is compared with its distribution over K shuffled-label runs (orange), in which y is permuted, $\mathbf{x}$ and $\mathbf{z}$ are kept, and the same pipeline is refitted. After Holm correction, a mechanism is flagged if C_m is significant, aligned if R_m and D_m are significant, and not aligned otherwise.

TABLE I
AI4I 2020 FAILURE MECHANISMS [15], THEIR TRIGGER CONDITIONS, EXPECTED FEATURES E_m, AND NUMBER OF FLAGGED OBSERVATIONS n. L/M/H ARE PRODUCT QUALITY TYPES.

Mech.	Trigger	E_m	n
TWF	tool wear at a random point in 200–240 min	tool wear	46
HDF	(process − air temp.) $<$ 8.6 K and speed $<$ 1380 rpm	both temps., their difference, speed	115
PWF	torque × speed (rad/s) outside [3500, 9000] W	torque, speed, power	95
OSF	tool wear × torque above 11,000/12,000/ 13,000 min·Nm for L/M/H	tool wear, torque, overstrain, its ratio	98
RNF	0.1% chance, independent of the process	–	19

a) Overlaps and label discrepancies: Twenty-four observations have several active flags, nine failures have none, and 18 of the 19 RNF observations are labeled non-failures. We train on the released labels, audit every observation where a mechanism is active, and repeat the audit on single-flag observations and with corrected labels (Section V-A). RNF has no trigger features and is examined only descriptively.

C. Feature Construction

Several rules combine sensor readings, so we construct four features: mechanical power $P = \tau(2\pi n/60)$, temperature difference $\Delta T = T_{\text{process}} - T_{\text{air}}$, overstrain $O = w\tau$, and overstrain ratio $Q = O/L(t)$, where τ, n, and w denote torque, rotational speed, and tool wear, respectively. The type-specific limit $L(t)$ is 11,000, 12,000, or 13,000 for product types `L`, `M`, and `H`.

They encode the interactions in the HDF, PWF, and OSF rules without using any labels, but they are correlated with the raw measurements, so we repeat the audit on the original features only (Section V-A).

D. Explanation Audit Statistics

We evaluate attribution using three complementary statistics. For feature j and observation set A, let

$$I_j(A) = \frac{1}{|A|}\sum_{i\in A}|\phi_j(x_i)|, \qquad s_j(A) = \frac{I_j(A)}{\sum_{k=1}^{d} I_k(A)} \tag{2}$$

denote the average absolute SHAP contribution and the attribution share of feature j, where $\phi_j(x_i)$ is computed by the cross-fitted model that did not train on observation i (Section III-E). We use absolute values because positive and negative contributions could otherwise cancel; the statistics thus measure how much a feature drives the prediction, and we check its direction separately (Section V-A). The fraction of attribution assigned to the expected trigger features of mechanism m is

$$r_m(A) = \sum_{j\in E_m} s_j(A). \tag{3}$$

From these quantities, we define three statistics:

$$R_m = r_m(S_m), \tag{4}$$
$$D_m = r_m(S_m) - r_m(F \setminus S_m), \tag{5}$$
$$C_m = \max_{j\notin E_m}\left[s_j(S_m) - s_j(F \setminus S_m)\right], \tag{6}$$

R_m measures how much attribution goes to the trigger features on mechanism m's observations, D_m whether that share is larger than on other failures, and C_m the largest mechanism-specific increase among features outside E_m. A large C_m raises concern about reliance on another feature but does not prove the feature causally irrelevant.

For example, on the 95 PWF observations, power, torque, and speed have mean $|\phi|$ of 4.96, 1.93, and 1.33 (Fig. 3a), out of 11.82 over all twelve inputs, so $R_{\mathrm{PWF}} \approx 0.70$.

E. Out-of-Sample Explanations

Explanations of training observations may reflect overfitting, so we use five-fold cross-fitting [41]: the full pipeline is fitted on four stratified folds and explains the fifth, giving every observation an explanation from a model that did not train on it. Section V-A contrasts this with in-sample SHAP.

F. Statistical Significance

A high score alone is not evidence of alignment: mechanism observations often have extreme sensor values that receive large SHAP values even when the model has learned nothing, which makes a comparison against random failure observations unreliable (Section V-B).

We instead build a *shuffled-label null*: we permute the overall failure labels, keeping sensors and mechanism flags, and repeat the full preprocessing, oversampling, training, cross-fitting, and explanation procedure with the hyperparameters selected on the original data. Because carrying these hyperparameters over could itself shape the null, we also rebuild it with the full hyperparameter search repeated on every permutation (Section V-A).

For an observed value T and null values $T^{(1)}, \ldots, T^{(K)}$, the upper-tail Monte Carlo p-value is

$$p = \frac{1 + \sum_{k=1}^{K} \mathbb{I}(T^{(k)} \geq T)}{K + 1}. \tag{7}$$

The primary audit uses $K = 1{,}000$, giving a minimum possible p-value of $1/1001 \approx 0.001$. We apply Holm's correction [42] across the four physical mechanisms for each statistic and use $\alpha = 0.05$. Since the smallest adjusted p-value is then $4/(K+1)$, $K \geq 80$ is needed to reach significance at all; in the primary audit, verdicts are identical at $K = 100$, 200, 500, and 1,000. The Monte Carlo standard error of a p-value near α, $\sqrt{\alpha(1-\alpha)/K}$, is 0.015 at $K = 200$ and 0.010 at $K = 500$, so we recommend $K \geq 200$, and $K \geq 500$ when an adjusted p lies within 0.03 of α. The null spread barely depends on mechanism size (log–log slope -0.14 over 40 mechanisms of the scaling runs), so the same K suits small and large mechanisms. Holm correction over all twelve R, D, and C tests also leaves every verdict unchanged. To complement the p-values, we report a null-standardized effect size $z = (T - \bar{T}_0)/s_0$, where $\bar{T}_0$ and s_0 are the mean and standard deviation of the null values, and 95% percentile bootstrap intervals for R_m and D_m obtained by resampling observations within S_m and $F \setminus S_m$ (5,000 resamples). The bootstrap holds the cross-fitted models fixed, so it reflects observation sampling only; variability from refitting is captured by the null.

We combine the three tests into one decision rule. A mechanism is *aligned* when both R_m and D_m are significantly larger than their null distributions and C_m is not; it is *flagged* when C_m is significant, whatever R_m and D_m show, and *not aligned* otherwise. The shortcut test can therefore overturn a positive verdict, which matters at large n, where a narrow null lets R_m pass even when a non-trigger feature takes much of the attribution (Section V-E). A flag names the responsible feature and its excess share, which should be inspected before acting on it. A non-significant shortcut result means that the audit did not detect such an increase; it does not establish that shortcuts are absent. RNF is excluded from the physical-alignment tests because it has no expected trigger features.

G. Model Training and Evaluation

We split the data into stratified training, validation, and test sets (70/20/10), fitting scaling and one-hot encoding on the training data. To handle the 3.4% failure rate, we compare SMOTE [43], Borderline-SMOTE [44], and ADASYN [45] by the validation F1 of a default XGBoost [46]. We compare seven classifiers: logistic regression, RBF-kernel SVM, k-NN ($k = 7$), a decision tree (depth 10), a random forest (300 trees, depth 12), XGBoost (300 trees, depth 6, learning rate 0.1), and an MLP (64 and 32 hidden units), with library defaults otherwise (scikit-learn 1.9.1, XGBoost 3.4.1, PyTorch 2.14.0; seed 42). The best model, XGBoost, is tuned by a 40-iteration randomized search with five-fold cross-validation on F1 (search space in the released code), oversampling inside each fold, and its threshold maximizes validation F1. We report failure-class precision, recall, F1, ROC-AUC, Brier score, and expected calibration error, with Clopper–Pearson intervals for precision and recall, 5,000 stratified bootstrap resamples otherwise, and five-fold cross-validation at threshold 0.5.

H. Experimental Design

The primary audit uses $K = 1{,}000$ and takes about 12 minutes on eight CPU cores; variants use $K = 200$ unless stated, the shortcut sweep $K = 500$.

a) Robustness checks: We repeat the audit with a null re-tuned on every permutation, four further seeds, single-mechanism observations, raw features only, in-sample SHAP, other label definitions and imbalance treatments, and a random forest. Interventional TreeSHAP uses 100, 500, or 1,000 background observations per fold, drawn from that fold's original (not oversampled) training data with up to three seeds, or, to examine TWF, only from non-failures, only from failures, or from the oversampled data. The MLP of Section III-G is explained on its logit with KernelSHAP [10] (25 background observations, 1,000 coalitions; also exact with 100 or 500), Integrated Gradients [47] (mean baseline, 1,024 steps), DeepSHAP, and SmoothGrad [48]. Dependence-aware variants sum path-dependent SHAP over four groups of related features (wear, thermal, load, product type) or compute exact

conditional Shapley values of these groups [23], estimated with Gaussian kernel weights over 2,000 training observations. Every variant uses a matching null.

b) Controlled shortcut experiment: To test whether the audit detects reliance on an unrelated but predictive feature, we introduce an artificial sensor

$$s_i = z_{i,\mathrm{HDF}} + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \sigma^2). \tag{8}$$

This noisy copy of the HDF flag has no physical role and is excluded from E_{HDF}; we decrease σ from 2 to 0.1 to make it increasingly predictive.

c) Scaling experiment: We generate new observations using the documented AI4I 2020 process [15], without copying released rows. The sensor distributions follow the documented process, and HDF, PWF, and OSF use their published rules. Tool wear increases until a lifetime drawn from $U(200, 240)$ min, where TWF occurs with probability 0.4. The generated failure rate is 3.5–4.0%, compared with 3.4% in the release.

We evaluate $n \in \{10^4, 3 \cdot 10^4, 10^5, 3 \cdot 10^5\}$ with three generation seeds for the first three sizes and one for the largest. Each setting is tested without a shortcut and with shortcut noise $\sigma \in \{0.35, 0.25\}$. The scaling experiments use $K = 200$ null models, and the generator is included in the released code.

d) Measured bearing data: We use the Case Western Reserve University bearing data [49]: 36 drive-end vibration recordings (12 kHz) with seeded faults at three locations (inner race, ball, outer race), three defect sizes, and four motor loads. Each location has a defect frequency fixed by the bearing geometry, so E_m holds the spectral power shares at that frequency and its second harmonic in the raw and envelope spectra (four of 19 features per 4,096-sample window; the rest are amplitude, shape, and broad-band statistics). A random forest (100 trees, depth 8) classifies the fault location, and each window is explained through the output of its own fault class. Cross-fitting holds out whole recordings (four stratified group folds), and the null permutes fault labels across recordings ($K = 1{,}000$). As a measured-data shortcut, we also scale each recording's amplitude by a fault-dependent gain (0.25, 1, and 4 for inner race, ball, and outer race), which changes only the log-RMS feature, outside every E_m.

e) Process data: We also use the Tennessee Eastman process in the DTU simulation set [50]: 400 complete 100-hour runs, 200 each of IDV4 (a step in the reactor cooling-water inlet temperature) and IDV5 (a step in the condenser cooling-water inlet temperature), injected at 30 h. From each run we take 16 samples before the fault (10–25 h) and 16 after (40–95 h) of all 53 variables; the label is fault activity and the mechanism the injected fault. E_m holds the three variables of the disturbed loop: reactor temperature, reactor coolant temperature, and reactor coolant flow for IDV4, and separator temperature, separator coolant temperature, and condenser coolant flow for IDV5. XGBoost (300 trees, depth 6) is cross-fitted over whole runs, and the null permutes labels within runs ($K = 200$).

TABLE II
FAILURE-CLASS TEST PERFORMANCE, SORTED BY F1. PARENTHESIZED INTERVALS INDICATE 95% BOOTSTRAP CONFIDENCE INTERVALS FOR F1.

Model	Prec.	Rec.	F1 (95% CI)	ROC-AUC
XGBoost (tuned)	**1.00**	0.79	**0.89** (0.79–0.95)	0.988
XGBoost	0.86	0.88	0.87 (0.78–0.94)	0.988
Random forest	0.79	0.88	0.83 (0.74–0.92)	**0.989**
MLP	0.65	**0.91**	0.76 (0.67–0.84)	0.988
Decision tree	0.68	0.82	0.75 (0.64–0.84)	0.927
SVM	0.42	0.88	0.57 (0.48–0.65)	0.976
k-NN	0.43	0.79	0.56 (0.46–0.65)	0.877
Logistic regression	0.24	0.85	0.37 (0.31–0.43)	0.943

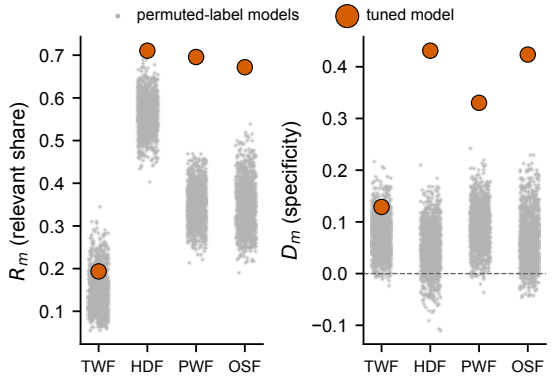


Fig. 2. Primary audit of the tuned XGBoost model. The observed relevant and specificity scores are compared with 1,000 shuffled-label models. HDF, PWF, and OSF exceed the null distributions on both measures, whereas TWF does not.

IV. Experimental Results

A. Predictive Performance

Borderline-SMOTE gives the highest validation F1 (0.826, versus 0.768 for SMOTE and 0.747 for ADASYN) and is used throughout. In Table II, tuned XGBoost reaches the highest test F1 (0.89 versus 0.87 untuned, threshold 0.978), although the confidence intervals overlap.

It finds 27 of the 34 test failures with no false alarm: precision 1.00 (95% CI 0.87–1.00), recall 0.79 (0.62–0.91), Brier score 0.0090 (0.033 for a base-rate predictor), and calibration error 0.008. Five-fold cross-validation gives failure-class F1 0.822 ± 0.013 and macro-F1 0.908 ± 0.007. This accuracy alone says nothing about whether the explanations match the failure mechanisms.

B. Alignment with Failure Mechanisms

Figure 2 and Table III compare the tuned XGBoost model's out-of-sample statistics with 1,000 shuffled-label models.

a) HDF, PWF, and OSF: All three mechanisms pass the relevant and specificity tests, with Holm-adjusted $p = 0.004$ for each. Their trigger features receive substantially more attribution on the corresponding mechanism observations than on other failures, with specificity scores between 0.33 and 0.43 and effect sizes of $z = 5.7$–8.4. The relevant-share effect is smallest for HDF ($z = 3.4$): its four trigger features

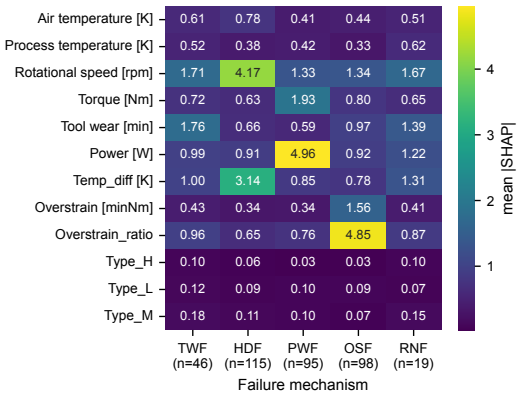


Fig. 3. Mean absolute SHAP of the tuned classifier by failure mechanism. Each mechanism draws on its own rule's features, a distinction that a global feature ranking hides.

include both temperatures and rotational speed, on which even shuffled-label models place much attribution (null mean 0.56). Bootstrap half-widths (0.02–0.03) are below the null standard deviations (0.04–0.06), so uncertainty comes mainly from model refitting. No shortcut is detected.

b) TWF: Tool wear failure does not pass either alignment test ($p_R = 0.201$, $p_D = 0.077$). Although tool wear is its most important feature, it receives only 19% of the attribution on TWF observations, compared with 15% under the null. Its rule draws a random failure point within a tool-wear interval, and only 46 observations carry the flag; Section V-D examines the verdict further.

C. Feature Attribution Across Mechanisms

In Fig. 3, rotational speed and temperature difference dominate HDF, power and torque PWF, and the overstrain features OSF, whereas a global SHAP ranking over all observations, as in prior AI4I work [16], hides these distinctions.

V. Experimental Analysis

Table IV summarizes how the audit responds to changes in the data, model, explainer, and null.

A. Robustness of Mechanism Alignment

a) Random seeds and overlapping mechanisms: HDF, PWF, and OSF pass both alignment tests across all five random seeds and when the analysis is restricted to observations with a single active mechanism. TWF fails both tests in every seed.

b) Re-tuned null: Repeating the hyperparameter search on each of 500 permutations widens the null (the search mostly selects shallow trees, and the 95th percentile of R_{OSF} rises from 0.45 to 0.61). The verdicts are unchanged, but 7 of 500 re-tuned null models reach the observed R_{OSF} (adjusted $p = 0.032$; 0.0498 with the first 200); the carried-over null is the less conservative reference.

c) In-sample versus out-of-sample explanations: In-sample SHAP passes all four mechanisms, including TWF ($p = 0.004$; R rises from 0.19 to 0.35), an optimistic verdict.

d) Original features only: Without the engineered features, HDF keeps its specificity but fails the relevant-share test, PWF and OSF stay aligned (OSF $p_D = 0.048$), and TWF still fails: only HDF's alignment depends on a derived feature, as a tree represents a temperature-difference rule poorly with separate temperatures.

e) Label anomalies and class imbalance: Removing the nine failures without a mechanism flag, or rebuilding the label as the OR of the five flags, changes no verdict (Table IV). Replacing Borderline-SMOTE with class weights keeps HDF, PWF, and OSF aligned and makes TWF specificity significant ($p_D = 0.020$, $p_R = 0.085$): without synthetic failures, path-dependent TreeSHAP moves toward the interventional verdict, as the reference-distribution account of Section V-D predicts. Borderline-SMOTE with 10 neighbours changes no verdict, with 3 it only makes TWF specificity significant ($p_D = 0.0498$), and a focal-loss objective without oversampling aligns all four mechanisms. With plain SMOTE or ADASYN, PWF and OSF stay aligned and HDF keeps its specificity ($p_D = 0.020$) but not its relevant share ($p_R = 0.109$), so the HDF share depends partly on the oversampler.

f) Misspecified trigger sets: Removing the derived features from E_m, or each mechanism's most attributed trigger, turns HDF, PWF, and OSF from aligned to flagged ($p_C = 0.020$), and each flag names the omitted feature (temperature difference or speed, power, overstrain ratio). An incomplete E_m therefore does not pass silently: the shortcut test points the expert to the missing feature.

g) Direction, removal, and dependence: On mechanism observations, 97–99% of the trigger features' attribution pushes toward failure, against 30–67% on other failures, so aligned attributions also point in the rule's direction. As a mechanism-level analogue of ROAR [34], retraining without a mechanism's raw trigger features cuts recall on its observations from 0.94 to 0.23 (HDF), 0.85 to 0.01 (PWF), and 0.95 to 0.23 (OSF), with smaller changes on other failures except under PWF removal (torque and speed also enter other rules); TWF recall is 0.07 even with all features. An attribution-free HSIC check [36] between each input and the cross-fitted prediction, on each mechanism's observations plus 1,000 normal ones, also ranks a trigger feature first for all four mechanisms (rank correlation with the SHAP shares 0.27–0.90). All three checks agree with the audit.

B. Why the Choice of Null Matters

An audit should reject a model that learned nothing. A classifier trained on shuffled labels fails our alignment tests, but compared with random failure observations under the same model, all four mechanisms appear significant (adjusted $p = 0.002$): the naive comparison approves a model trained on meaningless labels, because mechanism observations have unusual sensor values that receive large attributions regardless. Retraining under shuffled labels is therefore necessary.

C. Can the Audit Detect a Shortcut?

Figure 4 shows attribution shifting from the HDF trigger features to the planted sensor as its noise decreases.

TABLE III
PRIMARY AUDIT USING CROSS-FITTED SHAP VALUES (SEED 42, $K = 1{,}000$).

	n	Top feature	R_m [95% CI]	Null	z	p	D_m [95% CI]	Null	z	p	C_m	Null	p
TWF	46	tool wear	0.193 [0.17, 0.22]	0.152	0.9	0.201$^{\times}$	0.129 [0.10, 0.16]	0.070	1.5	0.077$^{\times}$	0.025	0.025	1.000
HDF	115	speed	0.711 [0.69, 0.73]	0.557	3.4	0.004	0.431 [0.41, 0.45]	0.049	8.4	0.004	0.001	0.015	1.000
PWF	95	power	0.696 [0.67, 0.72]	0.347	7.6	0.004	0.330 [0.30, 0.36]	0.088	5.7	0.004	0.001	0.011	1.000
OSF	98	overstrain ratio	0.672 [0.66, 0.69]	0.345	5.6	0.004	0.424 [0.40, 0.44]	0.067	7.0	0.004	-0.001	0.019	1.000
RNF	19	speed	0.186	0.240	–	0.985	–	–	–	–	–	–	–

Note: Null: mean over the shuffled-label models; z: (observed − null mean)/null s.d.; brackets: 95% bootstrap intervals with the models held fixed; p: Holm-adjusted over the four physical mechanisms; $^{\times}$: non-significant alignment test.

TABLE IV
HOLM-ADJUSTED AUDIT p-VALUES ACROSS EXPERIMENTAL VARIANTS AND FAILURE MECHANISMS.

		TWF			HDF			PWF			OSF		
Variant	K	R	D	C	R	D	C	R	D	C	R	D	C
Primary, seed 42	1000	0.201$^{\times}$	0.077$^{\times}$	1.000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000
single-mechanism rows	1000	0.187$^{\times}$	0.076$^{\times}$	1.000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000
null re-tuned per permutation	500	0.447$^{\times}$	0.359$^{\times}$	1.000	0.018	0.008	1.000	0.008	0.008	1.000	0.032	0.008	1.000
positive part of SHAP	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Seeds 1–4 (largest p)	200	0.269$^{\times}$	0.129$^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Raw features only	1000	0.345$^{\times}$	0.616$^{\times}$	1.000	0.190$^{\times}$	0.004	1.000	0.004	0.004	1.000	0.004	0.048	1.000
In-sample SHAP	1000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000
Unflagged failures removed	200	0.179$^{\times}$	0.075$^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Label = OR of flags	200	0.308$^{\times}$	0.144$^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Class weights, no oversampling	200	0.085$^{\times}$	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Focal loss, no oversampling	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Borderline-SMOTE, 3 neighbours	200	0.184$^{\times}$	0.0498	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
SMOTE (ADASYN alike)	200	0.169$^{\times}$	0.085$^{\times}$	1.000	0.109$^{\times}$	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
E_m without derived features	200	0.468$^{\times}$	0.209$^{\times}$	0.478	0.040	0.020	0.020†	0.179$^{\times}$	1.000$^{\times}$	0.020†	0.776$^{\times}$	1.000$^{\times}$	0.020†
E_m without top trigger	200	0.701$^{\times}$	0.209$^{\times}$	0.478	0.995$^{\times}$	0.020	0.020†	0.239$^{\times}$	0.935$^{\times}$	0.020†	0.995$^{\times}$	0.657$^{\times}$	0.020†
Interventional, background 100	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
failure-only background	200	0.995$^{\times}$	0.945$^{\times}$	0.020†	0.826$^{\times}$	0.149$^{\times}$	1.000	0.045	0.090$^{\times}$	1.000	0.040	0.020	1.000
non-failure background	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
oversampled background	200	0.045	0.020	0.896	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Pairwise (nearest normal reference)	200	0.020	0.020	0.498	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Random forest	200	0.617$^{\times}$	0.398$^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.040	0.020	1.000
Grouped TreeSHAP (4 groups)	200	0.816$^{\times}$	1.000$^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.030	1.000
Conditional group Shapley	200	0.020	0.020	1.000	0.020	0.040	0.517	0.020	0.020	1.000	0.020	0.040	1.000
MLP, KernelSHAP	200	0.975$^{\times}$	0.831$^{\times}$	0.975	0.388$^{\times}$	0.189$^{\times}$	1.000	0.358$^{\times}$	0.020	1.000	0.358$^{\times}$	0.075$^{\times}$	1.000
exact, background 100	200	1.000$^{\times}$	0.915$^{\times}$	0.756	0.557$^{\times}$	0.219$^{\times}$	0.965	0.557$^{\times}$	0.020	0.965	0.557$^{\times}$	0.149$^{\times}$	0.851
exact, background 500	100	0.990$^{\times}$	0.881$^{\times}$	0.792	0.683$^{\times}$	0.337$^{\times}$	1.000	0.683$^{\times}$	0.040	1.000	0.277$^{\times}$	0.089$^{\times}$	1.000
MLP, Integrated Gradients	200	0.338$^{\times}$	0.100$^{\times}$	0.522	0.020	0.020	1.000	0.0498	0.100$^{\times}$	0.478	0.030	0.020	1.000
MLP, DeepSHAP	200	0.920$^{\times}$	0.368$^{\times}$	0.478	0.020	0.020	1.000	0.104$^{\times}$	0.045	1.000	0.129$^{\times}$	0.045	1.000
MLP, SmoothGrad	200	0.970$^{\times}$	0.677$^{\times}$	0.040†	0.020	0.020	1.000	0.0498	0.020	1.000	0.020	0.020	1.000
Permuted labels (sanity)	200	1.000$^{\times}$	0.731$^{\times}$	1.000	0.199$^{\times}$	0.557$^{\times}$	1.000	1.000$^{\times}$	0.731$^{\times}$	1.000	1.000$^{\times}$	0.731$^{\times}$	0.637

Note: Holm-adjusted p-values of the R, D, and C tests; $^{\times}$: non-significant alignment test; †: significant shortcut test. The minimum attainable adjusted p-value is $4/(K+1)$. Interventional results use background seed 0; other seeds, backgrounds of 500 or 1,000, and raw features give the same verdicts. Indented interventional rows use a background of 100 observations drawn from the stated training subset.

For weak shortcuts ($\sigma \geq 0.4$, at most 8% of the attribution), HDF passes. At 13% ($\sigma = 0.35$) the relevant-share test still passes ($p = 0.044$), and it fails from 21% ($\sigma = 0.30$). The shortcut statistic becomes significant at about 35% ($\sigma \leq 0.25$), and the specificity test fails at about 42%. Test F1 stays between 0.84 and 0.92 throughout, so the model keeps good accuracy while increasingly relying on an unrelated feature. PWF and OSF pass in every setting. Weak shortcuts thus go undetected, and the shortcut statistic is less sensitive than the relevant-share test at moderate strengths; this sweep and the scaling study serve as empirical power curves. A normal approximation from the null and bootstrap spreads gives power near 1 for the R and D tests of HDF, PWF, and OSF and below 0.03 for TWF; at the first Holm step, an excess of 0.10–0.14 in attribution share over the null mean is detected with 80% power.

D. Effect of the Model and SHAP Method

HDF, PWF, and OSF pass the alignment tests using both path-dependent and interventional TreeSHAP, and their verdicts remain unchanged with a random forest.

a) Beyond trees: The audit needs only an attribution matrix, so it runs unchanged on the MLP (test F1 0.76) against 200 retrained MLP nulls (Table IV). With Integrated Gradients, HDF and OSF are aligned (adjusted $p \leq 0.030$), PWF passes the relevant-share test only ($p = 0.0498$; $p_D = 0.10$), and

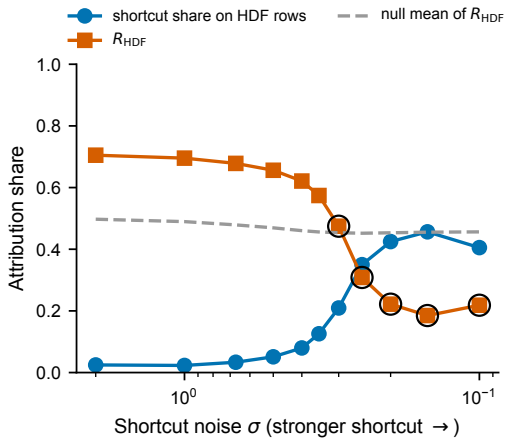


Fig. 4. Effect of an injected shortcut on HDF attribution. As noise decreases, the artificial sensor takes a larger share and the trigger features less. Markers indicate a failed relevant-share test.

TWF fails. With KernelSHAP (25-row background, 1,000 coalitions), no mechanism is aligned. Shuffled-label MLPs already place much attribution on the trigger features (R_{PWF} null mean 0.60–0.63 versus 0.35 for XGBoost), so raw shares of 0.47–0.67 do not exceed the null. Exact KernelSHAP (all 4,096 coalitions, 100- or 500-row background) changes the shares by at most 0.03 and gives the same verdicts, so neither sampling nor the background size is the cause. DeepSHAP and SmoothGrad behave like Integrated Gradients: both align HDF, SmoothGrad also PWF and OSF, DeepSHAP passes only the specificity test for PWF and OSF, and TWF fails (SmoothGrad flags overstrain, which is correlated with tool wear).

TWF behaves differently: it passes both tests with interventional TreeSHAP for every background size and seed ($R = 0.45$–0.47) but fails with path-dependent TreeSHAP and the random forest. Tool wear is correlated with the overstrain features overall (Spearman 0.92), and SHAP formulations distribute attribution among dependent features differently [23], but correlation does not explain the flip: on TWF observations it is only 0.13, and on the raw features alone TWF still fails path-dependent and passes interventional TreeSHAP ($R = 0.58$, $p = 0.020$).

The flip is better explained by the reference distribution. Path-dependent TreeSHAP weights paths by the training cover of the oversampled data, whereas our interventional background comes from the original training observations, about 97% non-failures. With three alternative backgrounds, each with a matching null (Table IV), a non-failure background reproduces the default ($R_{\mathrm{TWF}} = 0.46$); an oversampled background lowers it to 0.23, close to the path-dependent 0.19, though TWF still passes narrowly ($p_R = 0.045$); and a failure-only background lowers it to 0.08 and also breaks HDF. Against typical operation, tool wear distinguishes TWF observations; against other failures, it does not. Three further checks agree. A pairwise reference, the nearest normal-operation training observation of each explained observation [51], aligns TWF ($R = 0.31$, $p_R = p_D = 0.020$), as does a focal-loss model trained without oversampling ($R = 0.27$). Grouping correlated features does not help path-dependent SHAP (grouped $R_{\mathrm{TWF}} = 0.22$, null mean 0.27), but exact conditional Shapley values of the groups, measured against the training distribution, align all four mechanisms (TWF $R = 0.54$ versus 0.30, $p_R = p_D = 0.020$). Counting only the failure-pushing part of path-dependent SHAP also aligns TWF ($R = 0.67$): the model rarely predicts TWF observations as failures (recall 0.07), so their absolute attribution is dominated by features arguing against failure, while the evidence for failure is mostly tool wear. TWF thus fails only for absolute path-dependent attributions of the oversampled model; for audits asking why an observation is abnormal, we recommend a normal-operation reference, reported with the result.

E. Does the Audit Work with More Data?

On generated data with 10^4 to $3 \cdot 10^5$ observations and no injected shortcut, HDF, PWF, and OSF pass at every size and seed, and TWF still fails the relevant-share test, so more data alone does not resolve it. At $\sigma = 0.25$, the planted sensor takes 27–45% of HDF attribution. The relevant-share test detects this in all nine runs up to 10^5 observations but not at $3 \cdot 10^5$, where $R_{\mathrm{HDF}} = 0.472$ exceeds the null's 95th percentile of 0.417: as the null narrows, a model can beat shuffled-label models while relying heavily on a misleading feature. The shortcut score detects the strong shortcut in all ten runs, including at $3 \cdot 10^5$ (0.249 versus 0.211), so HDF is flagged rather than aligned there, the only case in which the shortcut test changes a verdict. The weaker shortcut ($\sigma = 0.35$, 6–20%) is not detected reliably.

The shortcut score also flags TWF in 7 of 30 runs from $3 \cdot 10^4$ observations onward, three without an injected sensor, always on air or process temperature with small excesses (0.03–0.05). The generator draws temperatures independently of tool life, so these are false flags made significant by the narrow null at large n; a flag should be read with its size and feature.

a) Computational cost: The audit requires $5(K+1)$ fits. One cross-fitted run takes 1.3, 31, and 115 seconds on one core for 10^4, $3 \cdot 10^5$, and 10^6 observations, roughly linear in n; on eight cores, a $K = 200$ audit takes about three minutes at 10^4 observations and one hour at $3 \cdot 10^5$; null runs parallelize.

F. Does the Audit Transfer to Measured and Process Data?

The forest identifies the fault location of all 36 held-out recordings (window accuracy 0.99). Table VI shows that inner- and outer-race attributions concentrate on their defect frequencies ($R_m = 0.44$ and 0.39, against null means of 0.23 and 0.15) and do so specifically, so both are aligned. Ball faults are not: their defect-frequency share (0.13) lies below the null mean (0.17), consistent with the benchmark finding that ball faults are the hardest to diagnose from these recordings [52]; the forest identifies them from other features. With the gain shortcut, every recording is still classified correctly, yet the outer-race output shifts toward log-RMS, whose excess share

TABLE V
AUDIT ON GENERATED DATA OF INCREASING SIZE ($K = 200$). SHARE: PLANTED SENSOR'S SHARE OF HDF ATTRIBUTION. COUNTS GIVE THE RUNS IN WHICH A TEST FIRES; MIN.: MINUTES PER AUDIT ON EIGHT CORES.

			HDF, planted sensor			TWF		
n	σ	Runs	Share	R fails	C flags	D passes	C flags	Min.
10,000	none	3	–	0/3	0/3	0/3	0/3	3
	0.35	3	0.158	2/3	0/3	0/3	0/3	3
	0.25	3	0.390	3/3	3/3	0/3	0/3	3
30,000	none	3	–	0/3	0/3	0/3	1/3	7
	0.35	3	0.128	0/3	0/3	2/3	0/3	6
	0.25	3	0.346	3/3	3/3	2/3	1/3	7
100,000	none	3	–	0/3	0/3	1/3	2/3	17
	0.35	3	0.100	0/3	0/3	2/3	1/3	19
	0.25	3	0.315	3/3	3/3	2/3	1/3	19
300,000	none	1	–	0/1	0/1	0/1	0/1	52
	0.35	1	0.063	0/1	0/1	0/1	0/1	59
	0.25	1	0.271	0/1	1/1	0/1	1/1	61

TABLE VI
AUDIT ON CWRU BEARING DATA ($K = 1{,}000$) AND TENNESSEE EASTMAN PROCESS DATA ($K = 200$); HOLM-ADJUSTED p.

Data	Fault	R_m	p	D_m	p	C_m	p	Verdict
CWRU	Inner	0.440	0.004	0.190	0.030	0.065	1.000	aligned
	Ball	0.133	$0.773^{\times}$	0.036	$0.211^{\times}$	0.051	1.000	not aligned
	Outer	0.392	0.003	0.131	0.027	0.042	1.000	aligned
+ gain	Inner	0.394	0.018	0.168	$0.054^{\times}$	0.063	0.989	not aligned
	Ball	0.105	$0.948^{\times}$	0.029	$0.547^{\times}$	0.020	0.989	not aligned
	Outer	0.269	0.018	0.050	$0.547^{\times}$	0.165	$0.006^{\dagger}$	flagged
TEP	IDV4	0.747	0.010	0.568	0.010	0.000	1.000	aligned
	IDV5	0.221	0.010	0.164	0.010	0.026	$0.010^{\dagger}$	flagged

is flagged ($C_m = 0.165$, $p = 0.006$), and both race faults lose specificity. As on AI4I 2020, perfect accuracy coexists with reliance on an acquisition artefact that the audit exposes.

On TEP, the classifier detects every IDV4 sample but only 39% of IDV5 samples (F1 0.76), as the controllers largely compensate for the condenser disturbance. IDV4 is aligned (75% of its attribution on the reactor cooling loop, null mean 6%). IDV5 attribution also concentrates on its loop (0.22 versus 0.06; top feature condenser coolant flow), but the shortcut test flags a small excess (0.026) on the E-feed valve, more likely a downstream control response missing from the three-variable map than a shortcut; a flag calls for inspection, not rejection.

G. Comparison with Simpler Audit Designs

A global SHAP ranking [16] gives no mechanism-level verdict; the naive null approves a shuffled-label model ($p = 0.002$, Section V-B); in-sample SHAP passes TWF ($p = 0.004$) and misses the strong shortcut at $3 \cdot 10^5$ observations ($p_C = 0.06$); and the relevant share alone also misses that shortcut (Section V-E). The full audit passes all three checks, although it cannot detect every misleading attribution.

VI. CONCLUSION

We proposed a statistical audit that compares SHAP attributions with known failure mechanisms and shuffled-label models. On AI4I 2020, a tuned XGBoost model (test F1 0.89) aligned with three mechanisms; tool wear depended on the explainer's reference distribution, and a neural network showed weaker, explainer-dependent alignment. On AI4I and CWRU data, accurate models relied on planted misleading features that simpler designs miss, so explanations must be evaluated beyond predictive accuracy.

For practitioners, our results suggest five rules: explain only held-out observations; compare with models retrained on shuffled labels, not random failure subsets; state the explainer's reference distribution, using normal operation to ask why an observation is abnormal; report effect sizes and flagged features with p-values, treating a mechanism as aligned only when R_m and D_m pass and C_m does not; and re-tune the null models when the budget allows.

VII. LIMITATIONS AND FUTURE WORK

The audit is evaluated on simulated AI4I 2020 and TEP data and on measured CWRU signals with seeded laboratory faults, so agreement is established with documented rules, process structure, and bearing kinematics rather than verified causes in operating plants. It needs expert-defined trigger features; an incomplete set is flagged rather than silently passed, but a flag cannot by itself distinguish a missing trigger from a shortcut. Verdicts depend on the explainer's reference distribution (TWF), on the oversampler (HDF's relevant share), and on the model (MLP); conditional SHAP was evaluated only for four feature groups, and LIME not at all. The shortcut statistic misses shortcuts below about a third of the attribution, power is approximated rather than derived and is low for TWF, OSF's relevant share is near the threshold under a re-tuned null ($p = 0.032$), and the bootstrap intervals hold the models fixed. Future work will apply the audit to operating-plant data with recorded failure causes, develop more sensitive shortcut statistics, and add rank- or distance-based statistics as complementary checks.

REFERENCES

[1] L. Cummins, A. Sommers, S. Bakhtiari Ramezani, S. Mittal, J. Jabour, M. Seale, and S. Rahimi, "Explainable predictive maintenance: A survey of current methods, challenges and opportunities," *IEEE Access*, vol. 12, pp. 57 574–57 602, 2024.

[2] Z. Allah Bukhsh, A. Saeed, I. Stipanovic, and A. G. Doree, "Predictive maintenance using tree-based classification techniques: A case of railway switches," *Transportation Research Part C: Emerging Technologies*, vol. 101, pp. 35–54, 2019.

[3] A. Jalali, A. Schindler, B. Haslhofer, and A. Rauber, "Machine learning interpretability techniques for outage prediction: A comparative study," *PHM Society European Conference*, vol. 5, no. 1, 2020.

[4] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you? Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 1135–1144.

[5] B. Steurtewagen and D. Van den Poel, "Adding interpretability to predictive maintenance by machine learning on sensor data," *Computers & Chemical Engineering*, vol. 152, p. 107381, 2021.

[6] S. Gawde, S. Patil, S. Kumar, P. Kamat, K. Kotecha, and S. Alfarhood, "Explainable predictive maintenance of rotating machines using LIME, SHAP, PDP, ICE," *IEEE Access*, vol. 12, pp. 29 345–29 361, 2024.

[7] M. L. Ndao, G. Youness, N. Niang, and G. Saporta, "Improving predictive maintenance: Evaluating the impact of preprocessing and model complexity on the effectiveness of eXplainable Artificial Intelligence methods," *Engineering Applications of Artificial Intelligence*, vol. 144, p. 110144, 2025.

[8] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020.

[9] S. Lapuschkin, S. Wäldchen, A. Binder, G. Montavon, W. Samek, and K.-R. Müller, "Unmasking Clever Hans predictors and assessing what machines really learn," *Nature Communications*, vol. 10, no. 1, p. 1096, 2019.

[10] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017.

[11] D. Janzing, L. Minorics, and P. Blöbaum, "Feature relevance quantification in explainable AI: A causal problem," in *Proc. 23rd Int. Conf. Artificial Intelligence and Statistics*, vol. 108, 2020, pp. 2907–2916.

[12] Y. Zhou, S. Booth, M. T. Ribeiro, and J. Shah, "Do feature attribution methods correctly attribute features?" in *Proc. AAAI Conf. Artificial Intelligence*, vol. 36, no. 9, 2022, pp. 9623–9633.

[13] A. Altmann, L. Tološi, O. Sander, and T. Lengauer, "Permutation importance: A corrected feature importance measure," *Bioinformatics*, vol. 26, no. 10, pp. 1340–1347, 2010.

[14] "AI4I 2020 Predictive Maintenance Dataset," UCI Machine Learning Repository, 2020, DOI: https://doi.org/10.24432/C5HS5C.

[15] S. Matzka, "Explainable artificial intelligence for predictive maintenance applications," in *2020 Third Int. Conf. Artificial Intelligence for Industries (AI4I)*, 2020, pp. 69–74.

[16] W. Zhao, J. Ding, X. Huang, and Y. Zhang, "Research on milling machine predictive maintenance based on machine learning and SHAP analysis in intelligent manufacturing environment," arXiv:2512.01205, 2025, accepted at ICEIEC 2025.

[17] O. Obeten and A. Jimoh, "Physics-informed, cost-aware fault classification with comparative explainability for predictive maintenance: a SHAP–LIME agreement analysis," *Scientific Reports*, 2026.

[18] S. Vollert, M. Atzmueller, and A. Theissler, "Interpretable machine learning: A brief survey from the predictive maintenance perspective," in *2021 26th IEEE Int. Conf. Emerging Technologies and Factory Automation (ETFA)*, 2021, pp. 1–8.

[19] I. Covert, S. Lundberg, and S.-I. Lee, "Explaining by removing: A unified framework for model explanation," *Journal of Machine Learning Research*, vol. 22, no. 209, pp. 1–90, 2021.

[20] M. Sundararajan and A. Najmi, "The many Shapley values for model explanation," in *Proc. 37th Int. Conf. Machine Learning*, vol. 119, 2020, pp. 9269–9278.

[21] H. Chen, I. C. Covert, S. M. Lundberg, and S.-I. Lee, "Algorithms to estimate Shapley value feature attributions," *Nature Machine Intelligence*, vol. 5, no. 6, pp. 590–601, 2023.

[22] H. Chen, J. D. Janizek, S. Lundberg, and S.-I. Lee, "True to the model or true to the data?" *arXiv preprint arXiv:2006.16234*, 2020.

[23] K. Aas, M. Jullum, and A. Løland, "Explaining individual predictions when features are dependent: More accurate approximations to Shapley values," *Artificial Intelligence*, vol. 298, p. 103502, 2021.

[24] I. E. Kumar, S. Venkatasubramanian, C. Scheidegger, and S. Friedler, "Problems with Shapley-value-based explanations as feature importance measures," in *Proc. 37th Int. Conf. Machine Learning*, vol. 119, 2020, pp. 5491–5500.

[25] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee, "From local explanations to global understanding with explainable AI for trees," *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020.

[26] P. Yu, A. Bifet, J. Read, and C. Xu, "Linear TreeShap," in *Adv. Neural Inf. Process. Syst.*, vol. 35, 2022.

[27] R. Wettenstein, R. Mitchell, and P. Yu, "Quadrature-TreeSHAP: Depth-independent TreeSHAP and Shapley interactions," arXiv:2605.04497, 2026.

[28] Y. Liu, S. Khandagale, C. White, and W. Neiswanger, "Synthetic benchmarks for scientific research in explainable machine learning," in *Proc. NeurIPS Datasets and Benchmarks Track*, 2021.

[29] J. R. Lee *et al.*, "XAI-Units: Benchmarking explainability methods with unit tests," in *Proc. ACM Conf. Fairness, Accountability, and Transparency (FAccT)*, 2025, arXiv:2506.01059.

[30] M. Yang and B. Kim, "Benchmarking attribution methods with relative feature importance," *arXiv preprint arXiv:1907.09701*, 2019.

[31] L. Arras, A. Osman, and W. Samek, "CLEVR-XAI: A benchmark dataset for the ground truth evaluation of neural network explanations," *Information Fusion*, vol. 81, pp. 14–40, 2022.

[32] C. Agarwal, S. Krishna, E. Saxena, M. Pawelczyk, N. Johnson, I. Puri, M. Zitnik, and H. Lakkaraju, "OpenXAI: Towards a transparent evaluation of model explanations," in *Adv. Neural Inf. Process. Syst. (Datasets and Benchmarks Track)*, vol. 35, 2022.

[33] Y. Li, Y. Zhang, K. Kawaguchi, A. Khakzar, B. Bischl, and M. Rezaei, "A dual-perspective approach to evaluating feature attribution methods," *Transactions on Machine Learning Research*, 2024.

[34] S. Hooker, D. Erhan, P.-J. Kindermans, and B. Kim, "A benchmark for interpretability methods in deep neural networks," in *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 9737–9748.

[35] Y. Rong, T. Leemann, V. Borisov, G. Kasneci, and E. Kasneci, "A consistent and efficient evaluation strategy for attribution methods," in *Proc. 39th Int. Conf. Machine Learning*, vol. 162, 2022, pp. 18 770–18 795.

[36] S. Da Veiga, "Global sensitivity analysis with dependence measures," *Journal of Statistical Computation and Simulation*, vol. 85, no. 7, pp. 1283–1305, 2015.

[37] M. Ojala and G. C. Garriga, "Permutation tests for studying classifier performance," *Journal of Machine Learning Research*, vol. 11, no. 62, pp. 1833–1863, 2010.

[38] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," in *Adv. Neural Inf. Process. Syst.*, vol. 31, 2018.

[39] P.-J. Kindermans, S. Hooker, J. Adebayo, M. Alber, K. T. Schütt, S. Dähne, D. Erhan, and B. Kim, "The (un)reliability of saliency methods," in *Explainable AI: Interpreting, Explaining and Visualizing Deep Learning*, 2019, vol. 11700, pp. 267–280.

[40] D. Slack, S. Hilgard, E. Jia, S. Singh, and H. Lakkaraju, "Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods," in *Proc. AAAI/ACM Conf. AI, Ethics, and Society*, 2020, pp. 180–186.

[41] V. Chernozhukov, D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins, "Double/debiased machine learning for treatment and structural parameters," *The Econometrics Journal*, vol. 21, no. 1, pp. C1–C68, 2018.

[42] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[43] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.

[44] H. Han, W.-Y. Wang, and B.-H. Mao, "Borderline-SMOTE: A new over-sampling method in imbalanced data sets learning," in *Int. Conf. Intelligent Computing*, 2005, pp. 878–887.

[45] H. He, Y. Bai, E. A. Garcia, and S. Li, "ADASYN: Adaptive synthetic sampling approach for imbalanced learning," in *2008 IEEE Int. Joint Conf. Neural Networks*, 2008, pp. 1322–1328.

[46] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[47] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *Proc. 34th Int. Conf. Machine Learning*, vol. 70, 2017, pp. 3319–3328.

[48] D. Smilkov, N. Thorat, B. Kim, F. Viégas, and M. Wattenberg, "SmoothGrad: Removing noise by adding noise," *arXiv preprint arXiv:1706.03825*, 2017.

[49] Case Western Reserve University, "Bearing data center: 12k drive end bearing fault data," https://engineering.case.edu/bearingdatacenter.

[50] C. C. Reinartz, M. Kulahci, and O. Ravn, "An extended Tennessee Eastman simulation dataset for fault-detection and decision support systems," *Computers & Chemical Engineering*, vol. 149, p. 107281, 2021.

[51] J. Xu, H. Chau, and A. Burden, "From abstract to actionable: Pairwise Shapley values for explainable AI," arXiv:2502.12525, 2025.

[52] W. A. Smith and R. B. Randall, "Rolling element bearing diagnostics using the Case Western Reserve University data: A benchmark study," *Mechanical Systems and Signal Processing*, vol. 64–65, pp. 100–131, 2015.